

# Grocery Delivery Analytics Pipeline

SYST52461 - Big Data Storage and Analysis | Term Project

## TEAM MEMBERS

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## Executive Summary

FreshRoute is a fictional grocery delivery company operating 25 stores across five Ontario cities. We built a deterministic PySpark pipeline that generates six related source tables, preserves realistic quality problems in Bronze, validates and standardizes them in Silver, and produces business-ready Gold tables for a native Databricks dashboard. The analysis covers revenue growth, category/store performance, discount response, delivery reliability, and customer repurchase behaviour.

The validated preview contains 3,540 completed orders, \$168,230 in net revenue, and a true order-level average order value of \$48.

## Data Model

| Table       | Role                          | Key relationships                  |
|-------------|-------------------------------|------------------------------------|
| customers   | Customer and loyalty profiles | CustomerID -> orders               |
| stores      | Store locations and formats   | StoreID -> products, orders        |
| products    | Store product catalogue       | ProductID -> order_items           |
| orders      | Order event and payment       | OrderID -> order_items, deliveries |
| order_items | Quantity, price, discount     | OrderID + ProductID                |
| deliveries  | Driver and fulfilment result  | OrderID -> orders                  |

Why this data is useful: grocery delivery links commercial decisions (assortment, price, discount and loyalty) to operational outcomes (distance, promised time and delay), allowing the dashboard to tell one connected business story.

# Medallion Architecture and Data Quality

| BRONZE                                                                                                                                                                    | SILVER                                                                                                                                                                   | GOLD                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Six raw Delta tables preserve source values exactly, including nulls, casing differences, multiple date/percentage formats, duplicates, invalid numbers, and orphan keys. | Column-specific rules standardize text, parse types, impute only defensible missing values, reject invalid records, and enforce primary/foreign keys through assertions. | Line-item, order, store, product, customer, monthly, discount, distance and loyalty tables expose stable business metrics for EDA and the dashboard. |

## Cleaning decisions

Customers: keep one row per ID, normalize city/loyalty/email, parse two date formats, reject ages outside 18-90, and median-impute age. Stores: constrain ratings to 1-5 and median-impute two invalid/missing values. Products: strip currency symbols, cast numbers, estimate missing cost at 65% of price, and reject nonpositive price or inventory. Orders: normalize status/payment, parse timestamps, and remove orphan customer/store keys. Order items: standardize `0.15`, `15`, and `15%` discounts, reject zero quantities/nonnumeric prices, and enforce product-store consistency. Deliveries: remove units, reject nonpositive distance/promised time, and calculate delay/on-time status only for delivered orders.

## Quality reconciliation

| Table       | Bronze | Silver | Removed |
|-------------|--------|--------|---------|
| customers   | 1,005  | 1,000  | 5       |
| stores      | 25     | 25     | 0       |
| products    | 250    | 244    | 6       |
| orders      | 4,000  | 3,879  | 121     |
| order_items | 12,002 | 11,226 | 776     |
| deliveries  | 4,000  | 3,872  | 128     |

## Validation and grain

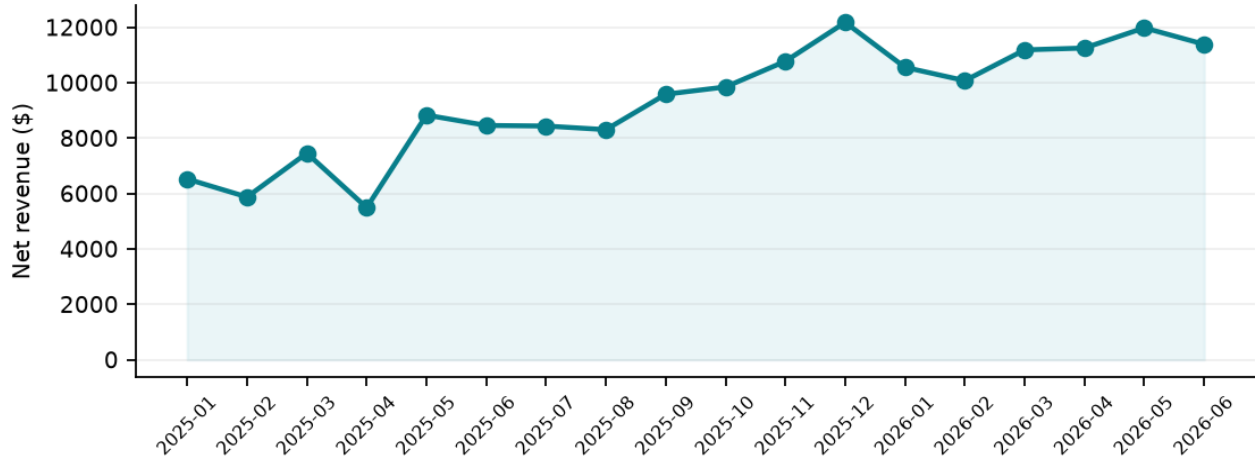
The Silver notebook asserts uniqueness for all six primary keys and zero orphan foreign keys. Gold revenue is calculated per valid completed line, then summed to exactly one row per order before average order value is computed. This avoids the common error of treating line-item revenue as order value. Dashboard datasets are generated only from validated Gold tables.

**Reproducibility: seed 52461 fixes all mock records, allowing every teammate and reviewer to reproduce the same row counts and findings.**

## EDA Findings I - Commercial Performance

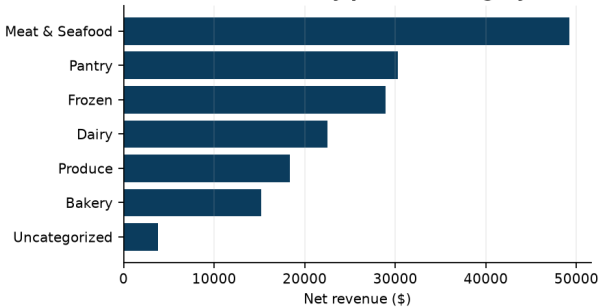
| Net revenue | Completed orders | Average order value | On-time delivery | Repeat customers |
|-------------|------------------|---------------------|------------------|------------------|
| \$168,230   | 3,540            | \$48                | 27.2%            | 82.8%            |

Monthly net revenue grows across the analysis period



Finding 1 - Demand strengthens over time. Monthly net revenue trends upward because the synthetic scenario combines moderate customer growth with higher weekend and December demand. The dashboard should retain monthly filters so managers can separate overall growth from seasonal peaks.

Net revenue by product category

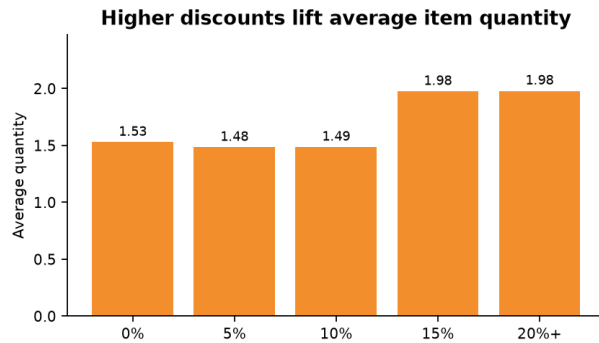


### Finding 2 - Category mix matters.

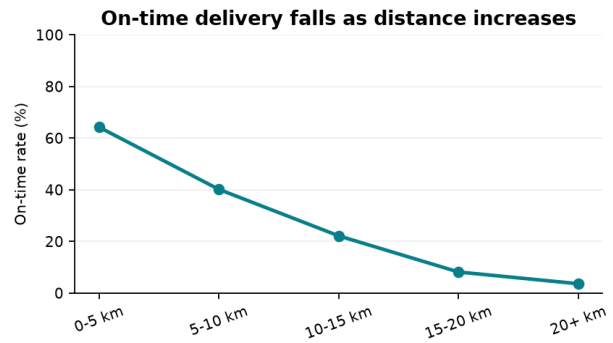
The leading category is **Meat & Seafood**, producing approximately **\$49,199** in net revenue. Managers should compare revenue with gross profit and units sold before expanding assortment; high revenue alone may reflect higher item prices.

**Store context:** FreshRoute Oakville 1 leads the store ranking at approximately \$9,217 in net revenue. The store dashboard pairs revenue with average order value and on-time rate so commercial success is not evaluated without operational quality.

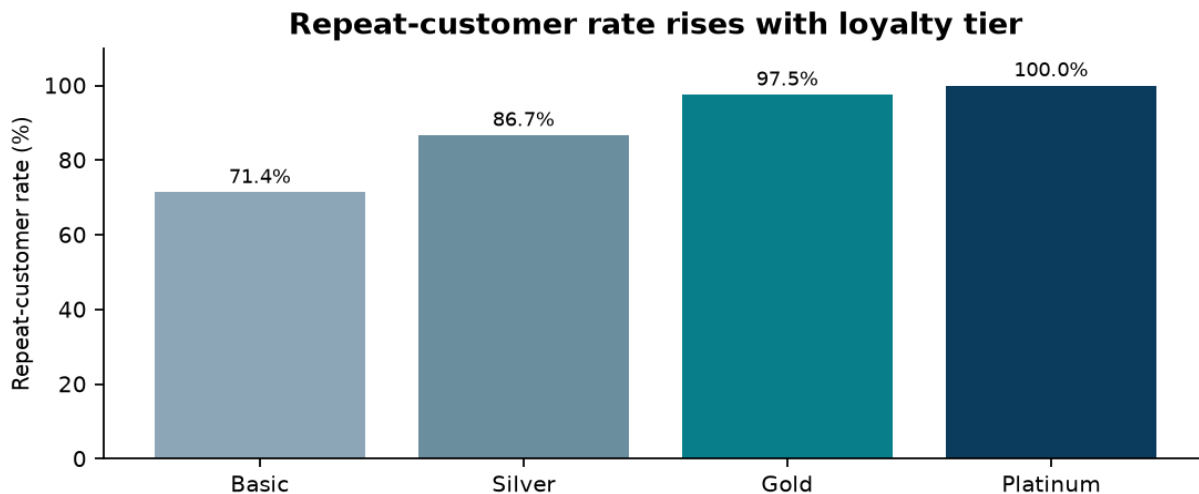
## EDA Findings II - Promotion, Delivery, and Loyalty



**Finding 3 - Discounts increase basket quantity.** Lines in the highest discount band average **1.98** units versus **1.53** with no discount. This supports selective promotion, but the dashboard also displays net revenue so volume is not mistaken for profitability.



**Finding 4 - Distance is the clearest delivery risk.** On-time performance declines from **64.3%** for the shortest band to **3.6%** for the longest. A practical response is distance-aware promises or smaller delivery zones during peak periods.



**Finding 5 - Loyalty tiers separate repurchase behaviour.** Gold and Platinum customers receive higher order-selection weights in the scenario and consequently show stronger repeat rates and order frequency. The finding demonstrates how a retailer could evaluate retention, but it is an engineered synthetic relationship and must not be generalized to real customers without observed data.

**Interpretation discipline: all findings describe the generated FreshRoute scenario. They validate that the pipeline and dashboard can expose useful relationships; they do not claim real-world causal effects.**

# Dashboard, Reflection, and Conclusion

## Dashboard story

The native Databricks dashboard opens with five KPI cards, then moves from company-level trend to category/store/product drivers, followed by promotional response, delivery risk, and loyalty behaviour. All visuals query persisted Gold tables. A YearMonth filter supports exploration without changing notebook code, and conclusion-oriented titles tell the audience why each visual matters.

| Section    | Visuals                            | Decision supported                       |
|------------|------------------------------------|------------------------------------------|
| Overview   | 5 KPI cards + monthly trend        | Monitor scale, growth and service        |
| Commercial | Category, top stores, top products | Assortment and store focus               |
| Promotion  | Discount vs. quantity/revenue      | Target discounts without ignoring margin |
| Operations | On-time rate and delay by distance | Promise times and delivery zones         |
| Customers  | Repeat rate and AOV by loyalty     | Retention and loyalty design             |

## Reflection and limitations

The most difficult design problem was coordinating six generators so foreign keys remained realistic while Bronze still contained testable defects. Explicit dependency order, one seed, and end-of-layer assertions made failures traceable. A second challenge was metric grain: revenue begins at line level, but average order value must be calculated after grouping to one order. With more time, we would add inventory events, weather and courier shifts, automate quality expectations, and compare synthetic distributions with a real grocery benchmark. Because the data is mocked, the dashboard proves analytical capability rather than real market performance.

## Team contribution plan

| Member           | Implemented and presents                             |
|------------------|------------------------------------------------------|
| Hazim Ali        | Customers, integration, repurchase KPI, introduction |
| Mannan           | Stores, store performance, store dashboard           |
| Maheshwar        | Products, category/product performance               |
| Sweta            | Orders, time/payment analysis                        |
| Omar Leopoldo    | Order items, revenue and discount analysis           |
| Shreyansh Pankaj | Deliveries, on-time analysis, reflection             |

## Conclusion

The project delivers a complete, reproducible medallion pipeline and a coherent dashboard story. It demonstrates that validated data modelling, correct aggregation grain and operational context are necessary to turn transactions into defensible business insights.

**GitHub repository:** [github.com/HazimAli07/grocery-delivery-analytics-pipeline](https://github.com/HazimAli07/grocery-delivery-analytics-pipeline)